FEDERICO COLOMBO

Computer Graphics Engineer • Technical Artist

Specialized in **real-time rendering**, low-level **engine architecture (C++)**, and **procedural pipelines**. Graduated with honors (**110L**) in Cinema & Media Engineering from Politecnico di Torino with active R&D in bio-adaptive volumetric SDF raymarching and mobile XR optimization.



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github/federicocolombo12

WORK EXPERIENCE

Technical Artist & Producer (XR Master Thesis R&D)
03/2026 - Present

Lemons in the Room • Firenze / Remote

- Architected a real-time bio-adaptive volumetric raymarching framework in HLSL for Meta Quest 3, rendering dynamic Signed Distance Fields (SDFs) driven by BLE biometric telemetry.
- Engineered GPU optimizations (Dynamic Bounding Volumes, adaptive ray stepping) locking **90 FPS on standalone XR hardware** (61.3% GPU time reduction, 0B GC alloc).
- Authored Morten Mikkelsen Surface Gradient shading on procedural volumes to eliminate triplanar UV seams, plus a sub-millisecond GPU Voxelizer for particle swarms.

Technical Artist, Gameplay Programmer & PM
11/2024 - 07/2026

LEVEL UP LAB (PoliTo Student Team) • Torino, Italy

- Authored custom shaders in HLSL (retro CRT curvature, water displacement) and programmed core C# gameplay for *Line Operator* & *Spicy Mercante*.
- Coordinated multidisciplinary pods of 6–7 student developers as Technical Producer and PM, enforcing Git LFS versioning and milestone sprints.

Teaching Assistant (GUI)
10/2024 - 01/2025

Politecnico di Torino • Torino, Italy

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created high-quality instructional content to improve comprehension.

Video Artist
09/2023 - 06/2024

RKH SRL • Torino, Italy

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.

EDUCATION

M.Sc. in Cinema & Media Engineering
2023 - 2026

Politecnico di Torino • 110/110 con Lode (Honors)

Focus: Real-Time Rendering, Game Engine Architecture, Computer Vision, Digital Signal Processing. Thesis: *PulseEngine — Volumetric Raymarching & Dynamic SDFs in XR*.

B.Sc. in Cinema & Media Engineering
2020 - 2023

Politecnico di Torino • Torino, Italy

Linear Algebra, Multivariate Calculus, Discrete Mathematics, Algorithms & Data Structures, C/C++ OOP.

CORE COMPETENCIES

Real-Time Rendering & Shaders

Compute shaders, Raymarching & SDFs, PBR equations, Deferred shading, SPOM, HLSL/GLSL.

Engine Architecture (C++)

Custom OpenGL engine development, memory layout, draw-call batching, multi-threading.

XR & Standalone Optimization

Meta Quest 3 (Snapdragon XR2), foveated raymarching, 90 FPS locked latency budgets.

Procedural DCC Pipelines

Blender Python API tooling, Unreal PCG & Niagara, automated asset pipelines.

SOFTWARE & TOOLS

C++ (17/20, STL)	Advanced
HLSL / GLSL	Compute/Ray
Unity 6 (URP, C#)	Expert
Unreal Engine 5	PCG, Niagara
Blender & Python API	Add-ons
Python 3	Pipeline
RenderDoc & Nsight	Profiling
Git & Git LFS	CI/VCS

LANGUAGES

Italian	Native Speaker
English	Professional Working (C1)